

UNIVERSITY NAME

*Analysis IV — Final Exam**Year: 2019***Exercise 1:**

Determine the nature (convergence/divergence) of the following sequences:

$$\begin{aligned} a) \ u_n &= \cos \left(\pi \frac{2n^3 + n^2 + an + b}{2n^2} \right), \quad a, b \in \mathbb{R} \\ b) \ u_n &= \left(1 - \frac{1}{n^a} \right)^n, \quad a > 0 \end{aligned}$$

Answer Area**Exercise 2:**

Consider the series of functions $\sum_{n \geq 1} f_n$ where

$$f_n(x) = \frac{x}{n(1 + n^2x)}.$$

Let $f(x) = \sum_{n=1}^{\infty} f_n(x)$ when it exists.

1. Show that f is defined and continuous on $\mathbb{R}^+ \setminus \{0\}$.
2. Show that there exists a unique $\lambda > 0$ such that $\lim_{x \rightarrow \infty} f(x) = \lambda$.
3. Show that f is differentiable on $\mathbb{R}_*^+$ and compute $f'(x)$.
4. Sketch the graph of f .

Answer Area**Exercise 3:**

Define the sequence (u_n) by $u_0 = u_1 = 1$ and for $n \in \mathbb{N}$:

$$u_{n+2} = u_{n+1} + 2u_n + (-1)^n.$$

Let $f(x) = \sum_{n=0}^{\infty} u_n x^n$ when it exists, and let R be its radius of convergence.

1. Show that $\forall n \in \mathbb{N}$,

$$1 \leq u_n \leq 2^{n+1} - 1$$

and deduce that $R \geq \frac{1}{2}$.

2. Show that for $|x| < \min(1, R)$, $f(x)$ satisfies:

$$f(x) = \frac{x^2 + x + 1}{(x + 1)^2(1 - 2x)}.$$

3. Give an explicit expression for u_n as a function of n .

Answer Area